



BONUS

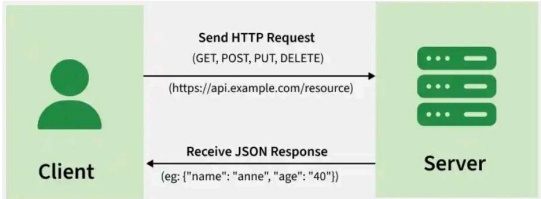


- **How passwords are stored in db ?**
 - Passwords are never stored as a plain text or encrypting in db., Instead they are transformed using a **cryptographic hashing called hashing combined of password with salting** which turns password to irreversible.
 - For eg.: 2 users having same password, by using basic hashing algorithm which will always produce the same exact hash for both. To prevent this use **salting**, when we create an account server will create a unique salt & attaches with password before hashing **resulting unique hash** & both users having same password also **looks completely different which makes password irreversible**.
 - Don't use MD5, SHA-1/256/512 hashing algorithm because they are fast & easily cracked by hackers. Instead use,
argon2id, bcrypt, scrypt
argon2id cryptographic hashing is a gold standard for secure hashing(uses salt + pepper)

- **Diff b/w SOAP & REST**

SOAP(Protocol)	REST(Architecture)
Strict rules	Flexible
Message format be only xml	json, xml, etc
Adds rule to Http[Secure, best for enterprise system - banking system,etc]	Can't add rule to http [Fast, best for mobile or web applications]

- **Stateless[No Memory]** → Stateless system does not store any information of previous requests.
eg.: ATM
- **Stateful[Has Memory]** → Stateful system stores session info(client state) of previous requests.
- **REST API ARCHITECTURE**: A REST API(Representational State Transfer API) enables communication b/w Client & Server over http. It exchanges data typically in JSON format using standard web protocol(http).
 - Uses http methods like GET, POST, PUT, PATCH & DELETE for CRUD Operations(CREATE, READ, UPDATE, DELETE) .
 - Clients sends request to server endpoints(URLs).
 - Server returns responses such as JSON, XML, etc.



- **Idempotent in REST API** : In REST API's an operation is called Idempotent if making the same request one or multiple times produces the same result in server.
eg.: Idempotent → GET, PUT, DELETE
Not Idempotent → POST, PATCH. partial update may change each time.

- **http v/s https :**

http	https (s-secure)
Hyper Text Transfer Protocol is the communication protocol which allows client to communicate with server .	secure version of http
No encryption	Encrypted to protect data, built on SSL/TLS security layer .
default port: 80	443

